Bounded drafting in repeated games

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1 Design

We compared direct action with a bounded draft-then-action procedure in two Nemotron models: Nano (30B-A3B) and Super (120B-A12B).[1] Players maximised expected own cumulative points in a repeated Prisoner's Dilemma: $CC = (3, 3)$, $CD = (0, 5)$, $DC = (5, 0)$, $DD = (1, 1)$. Choices were private until both players had acted.

Direct used one thinking-off action request (128-token cap). Draft used a thinking-on draft (1,024-token cap), then a separate thinking-off action request (128-token cap). Both received full completed history and used temperature 1 and top-$p = 1$. Only the current draft was added to the action request.

All 1,776 scheduled episodes completed in 24 balanced blocks:

- Track A: 1,536 episodes against disclosed AllC, AllD, Tit-for-Tat (TFT) or Grim policies. The 16 conditions covered continuation probabilities $\delta = 0.4, 0.6, 0.9$, one-round AllC/AllD, and ten-round TFT/Grim.
- Track B: 240 autonomous dyads covering all ten unordered pairings of the four model–procedure configurations, including self-play, at $\delta = 0.9$.

Four label/seat strata had six blocks each. Arms shared environment draws and stopping lengths within blocks; provider sampling randomness was not matched.

2 Scoring

Track A loss measures expected payoff forgone against a committed opponent. The gap for action a_t at state s_t is $g_t = V^*(s_t) - Q^*(s_t, a_t)$, where continuation after that action is optimal. Episode loss is

$$L_{\text{geometric}} = (1 - \delta) \sum_t g_t, \qquad L_{\text{fixed}} = \frac{1}{H} \sum_{t=1}^{H} g_t.$$

Geometric gaps are not discounted again: continuation is in Q^*, and termination supplies survival weighting. Lower loss is better; it measures neither action accuracy nor cooperation.

The primary contrast is draft minus direct mean loss within each model. The 16 conditions and four strata receive equal weight. Approximate 95% intervals use 20,000 stratified whole-block bootstrap draws.

3 Results

Each procedure has 384 Track A episodes. Drafting reduced average loss for both models (Table 1), but the benefit varied by opponent (Table 2). The opponent breakdown is exploratory.

[1] Hosted IDs: `nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B` and `nvidia/nemotron-3-super-120b-a12b`.

Model	Direct	Draft	Difference	Approx. 95% interval
Nano	0.6889	0.4942	−0.1947	[−0.3077, −0.0983]
Super	0.6150	0.3779	−0.2371	[−0.3095, −0.1735]

Table 1: Mean normalised oracle loss. Negative differences favour drafting.

Opponent	Nano difference	Super difference
AllC	−0.7562	−0.5146
AllD	−0.2135	−0.3708
Grim	−0.1575	−0.2542
TFT	+0.3485	+0.1913

Table 2: Draft minus direct loss, averaging four conditions per opponent.

Drafting worsened average performance against TFT. Against AllC and AllD, improvement means better selection of defection under the own-payoff objective, not more cooperation.

In Track B, mutual cooperation occurred in 189/2,060 rounds (9.2%); both defected in 1,042 rounds, and 829 were asymmetric. First-round mutual cooperation occurred in 26/240 dyads (10.8%). Rounds within a dyad are dependent; these counts do not supply 2,060 independent observations or a general ranking.

4 Interpretation and limitations

A best response to a commitment differs from an equilibrium [1]. Grim's initial cooperation threshold is $\delta = 1/2$; TFT's is $\delta = 2/3$. With ten announced rounds, either committed policy is best answered by nine cooperations then defection (32 own points). Finite-game backward induction for two strategic players does not imply immediate defection against these commitments. Geometric termination was hidden when choosing.

An earlier run stopped at 339/592 episodes after HTTP 502. Its outcomes were inspected before this successor, which retained the original 24-block schedule and excluded predecessor and engineering observations. These results are descriptive, not a new confirmatory preregistration.

Drafting changes compute, request count, thinking settings and conditioning together. Six blocks per stratum, one schedule, four policies and one provider window limit generalisation. Interval coverage and power are not established. There is no general model ranking or identified autonomous-play equilibrium; draft text does not verify internal beliefs.

5 Execution and reproduction

The run used 17,124 successful requests, with no retries or protocol failures, in approximately 3 hours 8 minutes. Estimated final inference cost was $4.9278, not a provider invoice.

Data and offline verification instructions are in `public/research/README.md`. Scores and intervals reproduce from revealed actions without inference. Raw responses, drafts and receipts remain private; public records cannot authenticate provider identity, raw transport or billing.

References

[1] Eric Rasmusen. *Games and Information: An Introduction to Game Theory*. Supplied fourth-edition draft, chapter headers dated 2005. Sections 1.2, 4.1, 5.1–5.2 and 6.5.